Open Book · Machine-Readable Build Specification

Claude Code Bootstrap

Paste-into-Claude-Code specification to rebuild the Open Book research-desk playground from an empty folder. Everything below is an instruction to the building agent. Give Claude Code this document plus an empty git repo and say: "Build this."

Mission statement for the building agent. You are building **Open Book**: a static, no-build-step advisory-desk simulator (four tabs: Solutions Views, Today's Focus, Advisor Book, Pre-Trade Analysis) in a J.P. Morgan Private Bank aesthetic, whose defining property is **transparent, derived, validated research discipline**: a goal-inference engine, a single idea→client fit engine, two conviction rubrics computed by a Python generator with hard validation, honest-dating via a fingerprint ledger, a tactical trigger gate, and sourced-or-stripped charts/technicals. Build files in the order of §2. Do not invent alternative architectures; where this spec gives a constant, use exactly that constant. Where a detail is unspecified, choose the simplest option consistent with §1.

1 · Non-negotiable principles

1. Idea→client links are **computed at render time** against the live client books — never hand-authored.
2. Every score decomposes into visible named components (axes with notes, pillars with one-line reads, goal drivers in plain English).
3. Conviction ("how good is the idea") and client-fit ("how right for this client") are separate engines, surfaced separately.
4. Facts are tagged `sourced` (≥2 independent sources) or `estimated`; validators warn, never silently fix.
5. Citation integrity: a named-author view citation requires a `basis` stating what was read and when; otherwise `ownView: true`. The generator rejects violations.

6. Dates are honest: `postedAt` moves only when an idea's argument-fingerprint changes (committed ledger).

7. Charts must carry fetched market data with provenance or they are stripped. Technicals come from a live scanner or stay tagged estimated.

8. Derivatives-first expression: every idea's `preferredExpression` is an options/structured/OTC construction; direct lines are alternates. MiFID gating decides who can trade what.

2 · File manifest (build in this order)

#	File	Contents
1	`styles.css`	Design system (§9)
2	`data.js`	`window.SEED`: themes, seed ideas, 8 clients, synthetic sector history, complexity taxonomy (§3)
3	`goals.js`	`window.GOALS`: deriveGoals + helpers (§4)
4	`mapping.js`	`window.MAPPING`: scoreIdeaForClient + flagClients + tradability (§5)
5	`scanner.js`	11-rule portfolio scan + thin facades over MAPPING
6	`expressions.js`	`window.EXPRESSIONS`: structure knowledge base, keyword resolver (§7)
7	`build_today_focus.py`	Generator/validator: rubrics, gates, ledger (§6)
8	`tv_technicals.py`, `fetch_chart_series.py`	Enrichment scripts (§6.5)
9	`today_focus.json`	First sweep payload (author via §8 procedure)
10	`email.js`, `charts.js`, `methodology.js`, `morgan.js`	Support engines
11	`app.js`, `openbook.js`, `index.html`, `portfolio.html`	Shell, tabs, drawers, modals, pre-trade

3 · data.js contract

3.1 Themes (9)

```
ai (AI & Productivity) · power (Power & Infrastructure) · duration (Core Fixed Income) broaden (Broadening Equity) · realassets (Real Assets) · resilience (Resilience & Protection) gold (Gold & Currency) · health (Healthcare Innovation) · structured (Structured Outcomes)
```

3.2 Seed idea object

```
{ id, themeId, intent: add|protect|trim|income, title, type: Thematic|Opportunistic|Strategic, assetClass, sector, bucket: Growth|Income|Preservation, conviction: "High"|"Medium-High"|"Medium", horizon, thesis, structures: [lead expression first — derivatives-first], riskProfile?: {vol, beta, structured}, tradeStatement?, tickers? }
```

3.3 Client object (8 clients)

```
{ id, name, ccy, aum ($m), classification: Retail|Professional, mifid, relationship, risk (free text — parsed), profile, split {assetClass: pct}, goals { objective, horizon ("Long-term · 7-10 yrs" — parsed), funding? { headline ("Grow to $75m by 2034" / "Fund $3.6m/yr..."), metricLabel, current, target, unit, status } }, liabilities?: [{name, amount, unit, note, rate?, near?}], positions: [{ name, ticker, assetClass, sector, ccy, weightPct, pnlPct, entryDate, entrySpot, mat, note, altKind?: uncorrelated|directional }], summary } INVARIANTS: Σ weightPct = 100 exactly; split ≡ Σ positions per assetClass.
```

3.4 The eight books (engineered test cases — reproduce these shapes)

id	Shape	Tests
aurora	Retail EUR €50m; MU 25.8% @ +295pnl; 72% USD; bond -15; gold+energy hedges; 10% cash	Concentration, FX mismatch, tax-harvest, Retail-OTC suppression
fable	Pro USD $38m; NVDA 24/AVGO 14/MU 13/ASML 19/MSFT 18; 6% autocall; 4% cash	Options-fluent monetisation (collars/overwrites)
scott	Retail $124m; 48% FI (one -24% long bond), dividend equity, PFE -38, O -11; 10% cash	Income mandate, bond swap, loss harvest, cash ladder

amar	Pro $71m; BTC 16% @ -35; gold 12 +118; infra 19; EM 12; macro HF 7	Preservation-bucket-full (gold gap-fit = 0), crypto rehab
jacob	Pro $96m; balanced; liabilities $19m mortgage + $2.4m Q3 tax; 5% cash; UNH -26	Liability-driven Income/Preservation push
prahnav	Retail $44m; NVDA 22 +279, CEG 15 +181; buckets on target	Goals-aligned top-down, Retail-collar problem bottom-up
morgan ("Mitch")	Pro $38m; NVDA 24, BTC 18, TSLA 11, 2x QQQ 10, venture 10; NO funding goal	Goal inference from stated-risk (aggressive) + book
tejpaul ("Fotis")	Retail $60m; balanced 57/35; no risk profile, no goals, no liabilities	Pure revealed inference → moderate

3.5 Synthetic sector history

```
client.sectorHistory[sector] = 24 monthly numbers, index 0 = current allocation EXACTLY. Shape from dominant position pnl: ≥50 → 'ramp' (start 12% of cur; index1 = cur×1.05); ≤-10 → 'fade' (start 130%); else 'core' (start 92%). Deterministic sine wobble ±4% on t≥4. Cash excluded. Label clearly as replaceable seed data.
```

3.6 Complexity taxonomy (order matters: structured checked BEFORE otc)

```
STRUCTURED_KEYWORDS = [structured note, structured re-entry, buffered, autocall, autocallable, acm+, phoenix, reverse convertible, revcon, fcn, range note, bren, certificate, capital-protected, capital protected, participation note, memory coupon, barrier note, credit-linked, twin-win, shark-fin, booster, halo] OTC_KEYWORDS = [collar, risk reversal, seagull, accumulator, decumulator, prepaid, variable forward, fx forward, forward / collar, dcd, dcs, dual currency, covered call, overwrite, cash-secured, call spread, protective put, strangle, straddle, fx put, fx call, fx option, put spread] complexityOf(s): structured → "structured" · otc → "otc" · else "non-complex" Retail may trade structured NOTES (packaged securities); Retail may NOT trade OTC.
```

4 · goals.js — deriveGoals(client)

```
PARAMS = { base:8, nearBillW:1.0, debtBallastW:0.3, concPivot:15, concW:1.0, drawdownBump:15,
shortHorizonW:3, incomeNeedW:8, serviceW:8, debtMatchW:0.6, reqReturnW:5, longHorizonW:3,
willingW:30, willingWNoGoal:45, cautionProtW:28, cautionIncW:18, revealRiskW:0.5,
revealDefenseW:0.3, revealConcW:0.2, fullDefenseAt:0.55, fullConcAt:0.25, serviceRate:0.06 }
PARSERS: horizon → {phase: drawdown if /drawdown|decumulat|retire/, years = midpoint of digits,
default 7} funding headline → incomeNeed ($m/yr via /[\$€£]?([\d.]+)m?\/yr/), terminalTarget (via
/to $Xm/), years (via /by 20xx/ − 2026), requiredReturn = (target/base)^(1/years) − 1 liabilities
→ nearBill (one-off: /tax|due|bridge|bill|payable|q[1-4]|settlement/), service =
debtLike(/mortgage|loan|line|credit|leverage|margin|borrow/) × rate (default 6%) WILLINGNESS:
stated — aggressive .95 · growth .80 · moderate .50 · conservative/income .30 · else null revealed
= .5×riskShare(Equity+Alts) + .3×(1 − min(1, defensive/.55)) + .2×min(1, topName%/25) blend =
stated==null ? revealed : .5 stated + .5 revealed RAW SCORES (each bucket = base + named terms):
Preservation = 8 + 1.0·nearBill% + 0.3·debtLoad% + 1.0·max(0, topName%−15) + 15·drawdown +
3·max(0,5−H) + 28·(1−willingness) Income = 8 + 8·incomeNeed% + 8·service% + 0.6·debtLoad% +
15·drawdown + 18·(1−willingness) Growth = 8 + 5·reqReturn + 3·max(0,H−6) + (reqReturn>0 ? 30 :
45)·willingness vector = largest-remainder normalise to 100. source/confidence: stated-goal .9 ·
stated-risk .7 · revealed .55. Expose components + plain-English drivers. goalOverride
short-circuits. FOLDING currentBuckets: Equity+directional Alts → Growth; FI/Credit/RealAssets →
Income; Cash/Commodity(gold)/uncorrelated Alts → Preservation; Structured by keyword purpose.
altKind explicit wins; else keyword-sniff (/uncorrelated|market-neutral|macro|multi-strat|hedge
fund| managed futures|trend|cta|relative value|absolute return|long\/short/ → Preservation).
goalAlignment(idea, client): gap = goalVector[bucketOfIdea] − currentBuckets[bucketOfIdea]; mult =
clamp(1 + gap/200, 0.85, 1.15).
```

5 · mapping.js — scoreIdeaForClient(idea, client)

```
PARAMS.weights (exact fractions): holdings .25/.85 ≈ .294 · mandate .25/.85 ≈ .294 · gap .20/.85 ≈
.235 · concSector .15/.85 ≈ .176 (Σ = 1.00) PARAMS.affinity: lambda .94; comfortByBucket
{Growth:25, Income:15, Preservation:12} (by the ASSET's bucket, per-sector overrides allowed);
penaltyPerPp 10, penaltyCap 100; noHolding 50. applyMin 45 · flagMin 50 · flagMax 6 · tiers ≥66
Strong / ≥48 Good / else Marginal. AXES (each 0-100, with a one-line note): 1 AFFINITY = max(0,
recency-weighted percentile (λ=.94) of current sector alloc within the 24-mo sectorHistory − max(0,
cur − sectorPeg)×10 capped 100) 2 MANDATE&RISK = .6×RiskSuitability(matrix
riskProfile{vol,beta,structured} × mandateClass) + .4×IntentFit(matrix: growth↔appreciation 90 ·
income↔yield 90 · preservation↔protection 92 · off-goal 40-75). mandateClass parsed from
client.risk. 3 GAP = max(0, (goalTarget[bucket] − currentBucket)/goalTarget × 100); 0 when bucket
full; fallback to sector headroom vs peg when no goal target. goalTarget = GOALS.goalsFor(client).
4 CONC-SECTOR: raw = (1 − HHI(normalised in-sector weights))×100; fitContribution = 100 − raw
(invertForFit=true, flippable); no in-sector holdings → 50. COMBINE: bracketFit = round(Σ
score×weight); alignedFit = bracketFit × goalAlignment.mult; tradability(idea,client) ∊ {0,1}: 0
iff Retail AND isOtc(naturalExpression); fit = trad × alignedFit; suppressed ideas surface WITH the
MiFID reason (never silently dropped). applies = trad && alignedFit ≥ 45; flagClients: fit ≥ 50,
top 6. Derived fallbacks: goalTypeOf(idea) from bucket/structures; riskProfileOf from sector beta
lists (HIGH_BETA: Tech/Crypto/Materials/Energy/Industrials/Consumer; LOW:
Utilities/Gold/Rates/Credit/ Infra/RealEstate/FX); naturalExpression = structures[0] if absent.
```

6 · build_today_focus.py — generator/validator

6.1 Payload shape

```
today_focus.json = { asOf, generated, sweep { sources[], rule, brief {summary, watch[3],
full[4-6]}, note }, earnings: [idea...], exEarnings: [idea...], meta {technicalsAsOf,
technicalsFeed} } Idea fields: id, direction (long|short|neutral), stance (earnings only:
pre-position|post-print), tradeStatement (REQUIRED one-sentence instrument+direction+view), intent,
action, preferredExpression, name, ticker, sector, assetClass, bucket, goalType, riskProfile,
naturalExpression, headline, thesis, themeId|null, offThemeWhy|null, ownView?, structures[],
variant {street, us, gap}, changeMyMind, conviction {pillars[]}, earnings {reportDate, reportWhen,
pre: impliedMovePct+impliedSource+historicalAvgMovePct+historicalSource+impliedVsHist / post:
result+resultSource+reaction+reactionSource, watch}, macro {metric, detail, source, watch}
(ex-earnings), facts [{text, tag: sourced|estimated}], sources [{name, kind:
data|news|calendar|analysis|view, basis (REQUIRED when kind=view)}], tactical?, trigger?,
triggered?, levels {tenor, entry, target, stop}?, chart {kind: spark|band, unit, band?, refs[],
caption, symbol, range}?
```

6.2 Conviction rubrics (computed, never authored)

```
EARNINGS (/8): asymmetry, consensus, catalyst, positioning — each 0-2, dq tag each. Bands: High =
raw≥6 & no zero & all sourced · "High — data gap" = same with ≥1 estimated · Medium = raw 4-5 or
any zero or consensus unverified · Low = raw<4 or 2+ zeros (excluded). EX-EARNINGS (/15): asymmetry
0-3, catalyst 0-2, consensus 0-2, thesis 0-2 (driver-specificity 0/1 + variant-view 0/1), houseview
0-2, positioning 0-2, technical 0-2. score = raw/15×100; High ≥75 · Medium ≥55 · Watch ≥0. GLOBAL
CAP: any pillar dq ∈ {estimated, unverified} → label capped at Medium (ex-earnings) / "High — data
gap" (earnings).
```

6.3 Validators (warn loudly, never silently pass)

- ≥2 sources per idea; every fact tagged; tradeStatement present; view-citations must carry basis.
- Stance-aware dates: pre-position report ≥ asOf; post-print report < asOf AND result present with resultSource="sourced".
- Coverage: FX sector present; Rates present (sector Rates or assetClass Fixed Income); both earnings stances present; ≥1 tactical idea (has levels) and ≥1 tactical FX.

- Tactical gate: levels ⇒ trigger text + boolean triggered required; only triggered ship; dropped ones reported "held back".

- Chart integrity: strip any chart lacking series (≥2 points) + seriesSource.

- Intent defaulting; pillar count/keys/max checks; unknown-pillar warnings.

6.4 Fingerprint ledger (today_focus.ledger.json)

```
fp = sha1(json of: tradeStatement, headline, thesis, changeMyMind, direction, intent, structures,
fact texts, pillar [key, score, authoredNote||note], levels, earnings.result) — excludes chart
series / indicator timestamps / computed totals. unseen id → postedAt = updatedAt = asOf · fp
unchanged → carry both · fp changed → keep postedAt, bump updatedAt. Warn when updatedAt > 10
days old. Ledger applied AFTER the tactical gate.
```

6.5 Enrichment scripts (stdlib only)

```
tv_technicals.py: POST scanner.tradingview.com/global/scan, columns [close, currency, SMA50,
SMA100, SMA200, RSI, BB.upper, BB.lower, Recommend.MA]. Rewrite ONLY the Technical pillar
(direction-aware: trend confirms long / vetoes short; neutral ideas scored on tape-calm), flip its
dq to sourced; add RSI colour to Positioning WITHOUT flipping its dq; rates yields skipped
honestly; failures leave the authored pillar untouched. Macro symbol map: DXY→TVC:DXY,
XAU→OANDA:XAUUSD, FX pairs→FX_IDC/OANDA, equities→NASDAQ:/NYSE:. fetch_chart_series.py: GET
query1.finance.yahoo.com/v8/finance/chart/{sym}?range={1mo|3mo|6mo|1y} &interval=1d → write series
(rounded closes), seriesAsOf, seriesSource ("Yahoo Finance daily closes · SYM · range to DATE").
PIPELINE: author payload → tv_technicals.py → fetch_chart_series.py → build_today_focus.py →
commit+push.
```

7 · expressions.js, email.js, scanner.js, methodology.js

- **expressions.js:** a dictionary of ~40 structures (zero-cost collar, call spread, put spread, risk reversal, strangle, covered call/overwrite, cash-secured puts, phoenix autocall, ACM+/HALO basket, reverse convertible, buffered note, capital-protected note, participation note, range accrual (BREN), dual-currency deposit, FX forward/collar, accumulator, bond ladder, gold accumulator, REIT/infra/utility baskets, liquid alternatives…). Each: label, complex flag, what/mechanics/example (realistic terms), pros[], cons[], goal-role scores {g,i,p}, context(client) personalisation. A keyword resolver maps free-text structure names → entries (substring matching, aliases). Every structures[] string used anywhere must resolve.

- **email.js:** idea × client → personalised letter: greeting, real-holding hook (name a position and its P&L), the desk view in client language, a concrete implementation block (structure, tenor, strikes, size), a tax/cash/FX action if the scanner flags one, a balanced risk paragraph, disclosure footer. One code path shared by all render surfaces.

- **scanner.js:** 11 severity-ranked findings rules over a book: single-name concentration >15%, underwater tax-harvest candidates, idle cash >8%, FX mismatch >30% non-base, missing protection with <5% Preservation vs goal, long-duration loss, liability-vs-cash squeeze, crypto drawdown, overwrite candidates (+50% winners), thin fixed income, missing income vs goal. Plus facades ideaFit/clientsForIdea delegating to MAPPING.

- **methodology.js:** builds `window.METHODOLOGY` by READING the live PARAMS objects (never copy constants) so the in-app "how scoring works" and the Morgan assistant cannot drift from code.

8 · The daily sweep procedure (operating loop)

1. Verify market state across five workstreams with ≥2 independent sources per fact: (a) Fed/rates/macro, (b) energy/gold/commodities, (c) FX/central banks, (d) US equities/earnings + a DISLOCATION SCAN (any widely-held large cap >15% off highs is a candidate), (e) global markets + the named-author list.

2. Named authors — core every sweep: Citrini, Serenity, TSCS, Brent Donnelly, Cem Karsan, Joseph Wang, SemiAnalysis. Conditional by domain: The Transcript (earnings), Benn Eifert/Moontower/Tier1 Alpha (vol), Fabricated Knowledge (semis), Rory Johnston/Doomberg (commodities), HighYield Harry (credit), The Market Ear/Brad Setser (flows), Andy Constan/Politano (macro). Cite ONLY what was actually read, with basis; otherwise ownView. Restated theses ≠ re-citation.

3. Re-underwrite the board: retire played/stale ideas (delete from payload), re-ground keepers (same id — the ledger preserves postedAt), add new ideas. Both earnings stances; FX + rates present; ≥1 triggered tactical with levels; every idea derivatives-first.

4. Write the sweep block: summary, 3 watch bullets, 4-6 full paragraphs, and a desk note archiving WHAT CHANGED (retired/new/re-grounded/gate decisions/dislocations scanned/author discipline/attribution line/discrepancies logged).

5. Run the three scripts; fix every validator warning; commit (`chore: market sweep refresh YYYY-MM-DD (one-line desk summary)`) and **push** (a local commit is invisible — the site deploys from the remote).

9 · Design system

```
Palette: ink #29211A on paper #FBF8F2 · bronze #9A7B4F · umber #5A3E2B · green #3F6B4E
(Preservation/positive) · slate #6E7E8C · red #8C3F3F · hairline #D8C9B2 · panel #F4ECE0. Type:
Georgia serif display+body; Helvetica small-caps letter-spaced kickers/tags/table heads; Consolas
code. Texture: hairline rules, double-rule breaks, uppercase 7.5pt tag chips, left-border callouts.
NO gradients/shadows/rounded cards — printed-research-note feel. Goal colours: Growth #29211A ·
Income #9A7B4F · Preservation #3F6B4E.
```

10 · Acceptance tests (run before calling it done)

1. Serve the folder; all four tabs render; no console errors.

2. Gold idea × Amar → Gap fit 0 (Preservation full); × Scott → high Gap fit.

3. A zero-cost-collar idea × Aurora (Retail) → fit 0, suppressed, MiFID reason SHOWN.

4. Mitch's derived goals ≈ Growth-dominant (~70%+) from stated-risk; Fotis ≈ moderate from revealed.

5. build_today_focus.py exits with ZERO warnings on the shipped payload; untriggered tacticals reported held back.

6. Re-running the build with no payload change moves NO dates (ledger no-op).

7. Editing one idea's thesis bumps only its updatedAt.

8. A chart block without a fetched series is stripped from today_focus.js.

9. Every structures[] string resolves in expressions.js (no dead expression links).

10. Conviction labels: any estimated pillar caps at Medium / "High — data gap".

What NOT to do. No build tooling, no frameworks, no npm — plain JS files loaded by script tags. No server-side rendering. Never hand-edit today_focus.js. Never fabricate a citation, a consensus number, or an earnings result — the validators treat these as defects, and so does the project's whole reason for existing.